PRAKTIKUM PEMROGRAMAN BERBASIS WEB (B)

TUGAS 6



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FAKULTAS TEKNIK
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Modifikasi index.php

Sebelum

```php
<?php
include "koneksi.php";

$query = mysqli_query($koneksi, "SELECT * FROM mahasiswa");
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <script src="https://cdn.jsdelivr.net/npm/@tailwindcss/browser@4"></script>
    <title>Data Mahasiswa</title>
</head>
<body class="bg-gray-50 p-6 font-sans text-gray-800">
    <div class="max-w-5xl mx-auto bg-white p-6 rounded-lg shadow-md">
        <h2 class="text-2xl font-bold mb-4 text-gray-800">Data Mahasiswa</h2>
        <a href="create.php"
        class="inline-block bg-blue-600 text-white font-semibold px-4 py-2 rounded mb-4">
            + Tambah Mahasiswa
        </a>

        <div class="overflow-x-auto">
            <table class="w-full border-collapse border border-gray-200 text-left">
                <thead class="bg-gray-100">
                    <tr>
                        <th class="border border-gray-200 p-3 font-semibold">NIM</th>
                        <th class="border border-gray-200 p-3 font-semibold">Nama</th>
                        <th class="border border-gray-200 p-3 font-semibold">Email</th>
                        <th class="border border-gray-200 p-3 font-semibold">Prodi</th>
                        <th class="border border-gray-200 p-3 font-semibold">
                            Angkatan
                        </th>
```

```php
        <th class="border border-gray-200 p-3 font-semibold">IPK</th>
        <th class="border border-gray-200 p-3 font-semibold text-center">
            Aksi
        </th>
    </tr>
</thead>

<tbody>
    <?php while ($data = mysqli_fetch_assoc($query)) { ?>
        <tr class="hover:bg-gray-50 transition">
            <td class="border border-gray-200 p-3"><?= $data['nim']; ?></td>
            <td class="border border-gray-200 p-3"><?= $data['nama']; ?></td>
            <td class="border border-gray-200 p-3"><?= $data['email']; ?></td>
            <td class="border border-gray-200 p-3"><?= $data['prodi']; ?></td>
            <td class="border border-gray-200 p-3"><?= $data['angkatan']; ?></td>
            <td class="border border-gray-200 p-3"><?= $data['ipk']; ?></td>
            <td class="border border-gray-200 p-3 text-center">
                <a href="edit.php?nim=<?= $data['nim']; ?>"
                    class="text-blue-600 hover:text-blue-800 hover:underline font-me
                    Edit
                </a>
                <span class="mx-1 text-gray-300">|</span>
                <a href="delete.php?nim=<?= $data['nim']; ?>"
                    onclick="return confirm('Yakin ingin menghapus data ini?')"
                    class="text-red-600 hover:text-red-800 hover:underline font-medi
                    Hapus
```

```
                                </td>
                            </tr>
                        <?php } ?>
                    </tbody>

                </table>
            </div>
        </div>
    </body>
    </html>
```

Sesudah

```php
<?php
require_once "koneksi.php";

$kataKunci = trim($_GET['cari'] ?? '');

$sql = "SELECT * FROM mahasiswa
        WHERE nim LIKE ?
           OR nama LIKE ?
           OR prodi LIKE ?
        ORDER BY nama ASC";

$stmt = mysqli_prepare($koneksi, $sql);
$kata = "%" . $kataKunci . "%";

mysqli_stmt_bind_param($stmt, "sss", $kata, $kata, $kata);
mysqli_stmt_execute($stmt);
$query = mysqli_stmt_get_result($stmt);

8 references
function e($nilai) {
    return htmlspecialchars((string)$nilai, ENT_QUOTES, 'UTF-8');
}
?>

<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <script src="https://cdn.jsdelivr.net/npm/@tailwindcss/browser@4"></script>
    <title>Data Mahasiswa</title>
</head>
```

```php
<body class="bg-gray-50 p-6 font-sans text-gray-800">
<div class="max-w-6xl mx-auto bg-white p-6 rounded-lg shadow-md">

    <h2 class="text-2xl font-bold mb-4">Data Mahasiswa</h2>

    <form method="GET" class="flex flex-wrap gap-2 mb-4">
        <input type="text" name="cari"
               value="<?= e($kataKunci) ?>"
               placeholder="Cari NIM, nama, atau prodi..."
               class="border rounded p-2 flex-1 min-w-48">

        <button class="bg-gray-700 text-white px-4 py-2 rounded">
            Cari
        </button>

        <a href="index.php"
           class="bg-gray-200 px-4 py-2 rounded">
            Reset
        </a>
    </form>

    <a href="create.php"
       class="inline-block bg-blue-600 text-white px-4 py-2 rounded mb-4">
        + Tambah Mahasiswa
    </a>

    <div class="overflow-x-auto">
        <table class="w-full border-collapse border border-gray-200 text-left">
            <thead class="bg-gray-100">
            <tr>
```

```php
    <th class="border p-3">NIM</th>
    <th class="border p-3">Nama</th>
    <th class="border p-3">Email</th>
    <th class="border p-3">No. HP</th>
    <th class="border p-3">Prodi</th>
    <th class="border p-3">Angkatan</th>
    <th class="border p-3">IPK</th>
    <th class="border p-3">Aksi</th>
</tr>
</thead>

<tbody>
<?php if (mysqli_num_rows($query) > 0): ?>
    <?php while ($data = mysqli_fetch_assoc($query)): ?>
    <tr class="hover:bg-gray-50">
        <td class="border p-3"><?= e($data['nim']) ?></td>
        <td class="border p-3"><?= e($data['nama']) ?></td>
        <td class="border p-3"><?= e($data['email']) ?></td>
        <td class="border p-3"><?= e($data['no_hp']) ?></td>
        <td class="border p-3"><?= e($data['prodi']) ?></td>
        <td class="border p-3"><?= e($data['angkatan']) ?></td>
        <td class="border p-3"><?= e($data['ipk']) ?></td>
        <td class="border p-3 whitespace-nowrap">
            <a class="text-blue-600 underline"
               href="edit.php?nim=<?= urlencode($data['nim']) ?>">
                Edit
            </a>

            <span class="mx-1">|</span>
```

```php
                <a class="text-red-600 underline"
                   href="delete.php?nim=<?= urlencode($data['nim']) ?>"
                   onclick="return confirm('Yakin ingin menghapus data ini?')">
                    Hapus
                </a>
            </td>
        </tr>
        <?php endwhile; ?>
    <?php else: ?>
        <tr>
            <td colspan="8" class="border p-4 text-center">
                Data mahasiswa tidak ditemukan.
            </td>
        </tr>
    <?php endif; ?>
    </tbody>
</table>
</div>

</div>
</body>
</html>

<?php
mysqli_stmt_close($stmt);
mysqli_close($koneksi);
?>
```

Modifikasi create.php

Sebelum

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <script src="https://cdn.jsdelivr.net/npm/@tailwindcss/browser@4"></script>
    <title>Tambah Mahasiswa</title>
</head>
<body class="bg-gray-50 p-6 font-sans text-gray-800">
    <form action="store.php" method="POST"
        class="max-w-sm mx-auto bg-white p-6 rounded-lg shadow-md flex flex-col">
        <h2 class="text-2xl font-bold mb-4 text-gray-800">
            Tambah Mahasiswa
        </h2>

        <label for="nim" class="text-sm font-semibold mb-1">NIM:</label>
        <input type="text" id="nim" name="nim" required
            class="mb-4 p-2 border border-gray-300 rounded">

        <label for="nama" class="text-sm font-semibold mb-1">Nama:</label>
        <input type="text" id="nama" name="nama" required
            class="mb-4 p-2 border border-gray-300 rounded">

        <label for="email" class="text-sm font-semibold mb-1">Email:</label>
        <input type="email" id="email" name="email" required
            class="mb-4 p-2 border border-gray-300 rounded">

        <label for="prodi" class="text-sm font-semibold mb-1">Prodi:</label>
        <input type="text" id="prodi" name="prodi" required
            class="mb-4 p-2 border border-gray-300 rounded">
```

```
        <label for="angkatan" class="text-sm font-semibold mb-1">Angkatan:</label>
        <input type="text" id="angkatan" name="angkatan" required
            class="mb-4 p-2 border border-gray-300 rounded">

        <label for="ipk" class="text-sm font-semibold mb-1">IPK:</label>
        <input type="number" id="ipk" name="ipk" step="0.01" required
            class="mb-6 p-2 border border-gray-300 rounded">

        <button type="submit" class="bg-blue-600 text-white font-bold py-2 rounded">
            Simpan
        </button>
    </form>
</body>
</html>
```

Sesudah

```
<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <script src="https://cdn.jsdelivr.net/npm/@tailwindcss/browser@4"></script>
    <title>Tambah Mahasiswa</title>
</head>

<body class="bg-gray-50 p-6 font-sans text-gray-800">
<form action="store.php" method="POST"
      class="max-w-lg mx-auto bg-white p-6 rounded-lg shadow-md flex flex-col">

    <h2 class="text-2xl font-bold mb-5">Tambah Mahasiswa</h2>

    <label for="nim" class="font-semibold mb-1">NIM</label>
    <input type="text" id="nim" name="nim" maxlength="15" required
           class="mb-4 p-2 border rounded">

    <label for="nama" class="font-semibold mb-1">Nama</label>
    <input type="text" id="nama" name="nama" maxlength="100" required
           class="mb-4 p-2 border rounded">

    <label for="email" class="font-semibold mb-1">Email</label>
    <input type="email" id="email" name="email" maxlength="120" required
           class="mb-4 p-2 border rounded">

    <label for="no_hp" class="font-semibold mb-1">Nomor HP</label>
    <input type="tel" id="no_hp" name="no_hp"
           pattern="[0-9+ -]{8,15}" maxlength="15" required
           title="Masukkan 8 sampai 15 karakter berupa angka, spasi, +, atau -"
           class="mb-4 p-2 border rounded">
```

```
    <label for="prodi" class="font-semibold mb-1">Program Studi</label>
    <input type="text" id="prodi" name="prodi" maxlength="80" required
           class="mb-4 p-2 border rounded">

    <label for="angkatan" class="font-semibold mb-1">Angkatan</label>
    <input type="number" id="angkatan" name="angkatan"
           min="1901" max="2155" required
           class="mb-4 p-2 border rounded">

    <label for="ipk" class="font-semibold mb-1">IPK</label>
    <input type="number" id="ipk" name="ipk"
           min="0" max="4" step="0.01" required
           placeholder="Contoh: 3.75"
           class="mb-6 p-2 border rounded">

    <button type="submit"
            class="bg-blue-600 hover:bg-blue-700 text-white font-bold py-2 rounded">
        Simpan Mahasiswa
    </button>

    <a href="index.php" class="text-center text-blue-600 mt-4">
        Kembali ke Daftar
    </a>
</form>
</body>
</html>
```

Modifikasi store.php

Sebelum

```php
<?php
include "koneksi.php";

if ($_SERVER["REQUEST_METHOD"] == "POST") {

    $nim = $_POST['nim'];
    $nama = $_POST['nama'];
    $email = $_POST['email'];
    $prodi = $_POST['prodi'];
    $angkatan = $_POST['angkatan'];
    $ipk = $_POST['ipk'];

    $query = mysqli_query(
        $koneksi,
        "INSERT INTO mahasiswa
        (nim, nama, email, prodi, angkatan, ipk)
        VALUES
        ('$nim', '$nama', '$email',
        '$prodi', '$angkatan', '$ipk')"
    );

    if ($query) {
        echo "<script>
                alert('Data mahasiswa berhasil disimpan');
                window.location.href = 'index.php';
              </script>";
    } else {
        echo "<script>
                alert('Data gagal disimpan');
                window.history.back();
              </script>";
    }
}
```

Sesudah

```php
<?php
require_once "koneksi.php";

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    header("Location: create.php");
    exit;
}

$nim = trim($_POST['nim'] ?? '');
$nama = trim($_POST['nama'] ?? '');
$email = trim($_POST['email'] ?? '');
$no_hp = trim($_POST['no_hp'] ?? '');
$prodi = trim($_POST['prodi'] ?? '');
$angkatanInput = $_POST['angkatan'] ?? '';
$ipkInput = $_POST['ipk'] ?? '';

if (
    $nim === '' || $nama === '' || $email === '' ||
    $no_hp === '' || $prodi === '' ||
    !filter_var($angkatanInput, FILTER_VALIDATE_INT) ||
    !is_numeric($ipkInput)
) {
    exit("Data tidak lengkap atau format tidak valid. <a href='create.php'>Kembali</a>");
}

$angkatan = (int)$angkatanInput;
$ipk = (float)$ipkInput;

if (
    strlen($nim) > 15 || strlen($nama) > 100 ||
    strlen($email) > 120 || strlen($prodi) > 80
) {
```

```
    exit("Panjang input melebihi batas. <a href='create.php'>Kembali</a>");
}

if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
    exit("Format email tidak valid. <a href='create.php'>Kembali</a>");
}

if (!preg_match('/^[0-9+ -]{8,15}$/', $no_hp)) {
    exit("Format nomor HP tidak valid. <a href='create.php'>Kembali</a>");
}

if ($angkatan < 1901 || $angkatan > 2155) {
    exit("Tahun angkatan tidak valid. <a href='create.php'>Kembali</a>");
}

if (!is_finite($ipk) || $ipk < 0 || $ipk > 4) {
    exit("IPK harus berada di antara 0 dan 4. <a href='create.php'>Kembali</a>");
}

$sql = "INSERT INTO mahasiswa
        (nim, nama, email, no_hp, prodi, angkatan, ipk)
        VALUES (?, ?, ?, ?, ?, ?, ?)";

$stmt = mysqli_prepare($koneksi, $sql);

if (!$stmt) {
    exit("Gagal menyiapkan query.");
}
```

```
mysqli_stmt_bind_param(
    $stmt,
    "sssssid",
    $nim, $nama, $email, $no_hp, $prodi, $angkatan, $ipk
);

if (mysqli_stmt_execute($stmt)) {
    mysqli_stmt_close($stmt);
    mysqli_close($koneksi);

    echo "<script>
        alert('Data mahasiswa berhasil disimpan');
        window.location.href = 'index.php';
    </script>";
    exit;
}

$nomorError = mysqli_stmt_errno($stmt);
mysqli_stmt_close($stmt);
mysqli_close($koneksi);

if ($nomorError === 1062) {
    exit("NIM atau email sudah terdaftar. Gunakan data yang berbeda. <a href='create.php'>Kembali</a
}

exit("Data gagal disimpan. Periksa struktur tabel dan koneksi database.");
?>
```

Modifikasi edit.php

Sebelum

```
<?php
include "koneksi.php";

$nim = $_GET['nim'];

$query = mysqli_query(
        $koneksi,
        "SELECT * FROM mahasiswa WHERE nim = '$nim'"
    );

$data = mysqli_fetch_assoc($query);
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <script src="https://cdn.jsdelivr.net/npm/@tailwindcss/browser@4"></script>
    <title>Ubah Data <?= $data['nama'] ?></title>
</head>
<body class="bg-gray-50 p-6 font-sans text-gray-800">
    <form action="update.php" method="POST"
        class="max-w-sm mx-auto bg-white p-6 rounded-lg shadow-md flex flex-col">
        <h2 class="text-2xl font-bold mb-4 text-gray-800">
            Ubah Data <?= $data['nama'] ?>
        </h2>

        <label for="nim" class="text-sm font-semibold mb-1">NIM:</label>
        <input type="text" id="nim" name="nim" value="<?= $data['nim'] ?>" required
            class="mb-4 p-2 border border-gray-300 rounded">
```

```
        <label for="nama" class="text-sm font-semibold mb-1">Nama:</label>
        <input type="text" id="nama" name="nama" value="<?= $data['nama'] ?>" required
            class="mb-4 p-2 border border-gray-300 rounded">

        <label for="email" class="text-sm font-semibold mb-1">Email:</label>
        <input type="email" id="email" name="email" value="<?= $data['email'] ?>" required
            class="mb-4 p-2 border border-gray-300 rounded">

        <label for="prodi" class="text-sm font-semibold mb-1">Prodi:</label>
        <input type="text" id="prodi" name="prodi" value="<?= $data['prodi'] ?>" required
            class="mb-4 p-2 border border-gray-300 rounded">

        <label for="angkatan" class="text-sm font-semibold mb-1">Angkatan:</label>
        <input type="text" id="angkatan" name="angkatan" value="<?= $data['angkatan'] ?>" required
            class="mb-4 p-2 border border-gray-300 rounded">

        <label for="ipk" class="text-sm font-semibold mb-1">IPK:</label>
        <input type="number" id="ipk" name="ipk" value="<?= $data['ipk'] ?>" step="0.01" required
            class="mb-6 p-2 border border-gray-300 rounded">

        <button type="submit" class="bg-blue-600 text-white font-bold py-2 rounded">
            Simpan
        </button>
    </form>
</body>
</html>
```

Sesudah

```
<?php
require_once "koneksi.php";

$nim = trim($_GET['nim'] ?? '');

if ($nim === '') {
    exit("NIM tidak diberikan. <a href='index.php'>Kembali</a>");
}

$stmt = mysqli_prepare(
    $koneksi,
    "SELECT * FROM mahasiswa WHERE nim = ?"
);
mysqli_stmt_bind_param($stmt, "s", $nim);
mysqli_stmt_execute($stmt);

$result = mysqli_stmt_get_result($stmt);
$data = mysqli_fetch_assoc($result);

mysqli_stmt_close($stmt);

if (!$data) {
    mysqli_close($koneksi);
    exit("Data mahasiswa tidak ditemukan. <a href='index.php'>Kembali</a>");
}

15 references
function e($nilai) {
    return htmlspecialchars((string)$nilai, ENT_QUOTES, 'UTF-8');
}

mysqli_close($koneksi);
?>
```

```
<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <script src="https://cdn.jsdelivr.net/npm/@tailwindcss/browser@4"></script>
    <title>Edit Mahasiswa</title>
</head>

<body class="bg-gray-50 p-6 font-sans text-gray-800">
<form action="update.php" method="POST"
      class="max-w-lg mx-auto bg-white p-6 rounded-lg shadow-md flex flex-col">

    <h2 class="text-2xl font-bold mb-5">Edit Data Mahasiswa</h2>

    <label for="nim" class="font-semibold mb-1">NIM</label>
    <input type="text" id="nim" name="nim"
           value="<?= e($data['nim']) ?>" readonly
           class="mb-4 p-2 border rounded bg-gray-100">

    <label for="nama" class="font-semibold mb-1">Nama</label>
    <input type="text" id="nama" name="nama" maxlength="100"
           value="<?= e($data['nama']) ?>" required
           class="mb-4 p-2 border rounded">

    <label for="email" class="font-semibold mb-1">Email</label>
    <input type="email" id="email" name="email" maxlength="120"
           value="<?= e($data['email']) ?>" required
           class="mb-4 p-2 border rounded">

    <label for="no_hp" class="font-semibold mb-1">Nomor HP</label>
    <input type="tel" id="no_hp" name="no_hp" maxlength="15"
           pattern="[0-9+ -]{8,15}"
```

```
           value="<?= e($data['no_hp']) ?>" required
           class="mb-4 p-2 border rounded">

    <label for="prodi" class="font-semibold mb-1">Program Studi</label>
    <input type="text" id="prodi" name="prodi" maxlength="80"
           value="<?= e($data['prodi']) ?>" required
           class="mb-4 p-2 border rounded">

    <label for="angkatan" class="font-semibold mb-1">Angkatan</label>
    <input type="number" id="angkatan" name="angkatan"
           min="1901" max="2155"
           value="<?= e($data['angkatan']) ?>" required
           class="mb-4 p-2 border rounded">

    <label for="ipk" class="font-semibold mb-1">IPK</label>
    <input type="number" id="ipk" name="ipk"
           min="0" max="4" step="0.01"
           value="<?= e($data['ipk']) ?>" required
           class="mb-6 p-2 border rounded">

    <button type="submit"
            class="bg-blue-600 hover:bg-blue-700 text-white font-bold py-2 rounded">
        Simpan Perubahan
    </button>

    <a href="index.php" class="text-center text-blue-600 mt-4">
        Batal
    </a>
</form>
</body>
</html>
```

Modifikasi update.php

Sebelum

```php
<?php
include "koneksi.php";

$nim = $_POST['nim'];
$nama = $_POST['nama'];
$email = $_POST['email'];
$prodi = $_POST['prodi'];
$angkatan = $_POST['angkatan'];
$ipk = $_POST['ipk'];

$query = mysqli_query(
    $koneksi,
    "UPDATE mahasiswa
    SET nama='$nama',
        email='$email',
        prodi='$prodi',
        angkatan='$angkatan',
        ipk='$ipk'
    WHERE nim='$nim'"
);

if ($query) {
    echo "<script>
        alert('Data mahasiswa berhasil diubah');
        window.location.href = 'index.php';
        </script>";
} else {
echo "<script>
        alert('Data gagal diubah');
        window.history.back();
        </script>";
}
?>
```

Sesudah

```php
<?php
require_once "koneksi.php";

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    header("Location: index.php");
    exit;
}

$nim = trim($_POST['nim'] ?? '');
$nama = trim($_POST['nama'] ?? '');
$email = trim($_POST['email'] ?? '');
$no_hp = trim($_POST['no_hp'] ?? '');
$prodi = trim($_POST['prodi'] ?? '');
$angkatanInput = $_POST['angkatan'] ?? '';
$ipkInput = $_POST['ipk'] ?? '';

if (
    $nim === '' || $nama === '' || $email === '' ||
    $no_hp === '' || $prodi === '' ||
    !filter_var($angkatanInput, FILTER_VALIDATE_INT) ||
    !is_numeric($ipkInput)
) {
    exit("Data tidak lengkap atau format tidak valid. <a href='index.php'>Kembali</a>");
}

$angkatan = (int)$angkatanInput;
$ipk = (float)$ipkInput;

if (
    strlen($nim) > 15 || strlen($nama) > 100 ||
    strlen($email) > 120 || strlen($prodi) > 80
) {
    exit("Panjang input melebihi batas.");
}
```

```php
}

if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
    exit("Format email tidak valid.");
}

if (!preg_match('/^[0-9+ -]{8,15}$/', $no_hp)) {
    exit("Format nomor HP tidak valid.");
}

if ($angkatan < 1901 || $angkatan > 2155) {
    exit("Tahun angkatan tidak valid.");
}

if (!is_finite($ipk) || $ipk < 0 || $ipk > 4) {
    exit("IPK harus berada di antara 0 dan 4.");
}

$cek = mysqli_prepare(
    $koneksi,
    "SELECT nim FROM mahasiswa WHERE email = ? AND nim <> ?"
);
mysqli_stmt_bind_param($cek, "ss", $email, $nim);
mysqli_stmt_execute($cek);
mysqli_stmt_store_result($cek);

if (mysqli_stmt_num_rows($cek) > 0) {
    mysqli_stmt_close($cek);
    mysqli_close($koneksi);
    exit("Email sudah digunakan mahasiswa lain.");
}
mysqli_stmt_close($cek);
```

```php
$sql = "UPDATE mahasiswa
        SET nama = ?, email = ?, no_hp = ?, prodi = ?,
            angkatan = ?, ipk = ?
        WHERE nim = ?";

$stmt = mysqli_prepare($koneksi, $sql);
mysqli_stmt_bind_param(
    $stmt,
    "ssssids",
    $nama, $email, $no_hp, $prodi, $angkatan, $ipk, $nim
);

$berhasil = mysqli_stmt_execute($stmt);
$nomorError = mysqli_stmt_errno($stmt);

mysqli_stmt_close($stmt);
mysqli_close($koneksi);

if ($berhasil) {
    echo "<script>
        alert('Data mahasiswa berhasil diubah');
        window.location.href = 'index.php';
    </script>";
} elseif ($nomorError === 1062) {
    exit("Email sudah digunakan mahasiswa lain.");
} else {
    exit("Data gagal diubah. Periksa kembali database.");
}
?>
```

Modifikasi delete.php

Sebelum

```php
<?php
include "koneksi.php";

$nim = $_GET['nim'];

$query = mysqli_query(
    $koneksi,
    "DELETE FROM mahasiswa
    WHERE nim='$nim'"
);

if ($query) {
    echo "<script>
        alert('Data mahasiswa berhasil dihapus');
        window.location.href = 'index.php';
        </script>";
} else {
    echo "<script>
        alert('Data gagal dihapus');
        window.history.back();
        </script>";
}
?>
```

Sesudah

```php
<?php
require_once "koneksi.php";

$nim = trim($_GET['nim'] ?? '');

if ($nim === '') {
    mysqli_close($koneksi);
    exit("NIM tidak diberikan. <a href='index.php'>Kembali</a>");
}

$stmt = mysqli_prepare(
    $koneksi,
    "DELETE FROM mahasiswa WHERE nim = ?"
);

mysqli_stmt_bind_param($stmt, "s", $nim);
mysqli_stmt_execute($stmt);

$jumlahDihapus = mysqli_stmt_affected_rows($stmt);

mysqli_stmt_close($stmt);
mysqli_close($koneksi);
```

```php
if ($jumlahDihapus > 0) {
    echo "<script>
        alert('Data mahasiswa berhasil dihapus');
        window.location.href = 'index.php';
    </script>";
} else {
    echo "<script>
        alert('Data tidak ditemukan atau sudah dihapus');
        window.location.href = 'index.php';
    </script>";
}
?>
```

lima bagian kode yang paling penting.

1. Menampilkan data — index.php
   $query = mysqli_stmt_get_result($stmt);
   Digunakan untuk mendapatkan hasil query pencarian mahasiswa dari database. Hasilnya kemudian ditampilkan dalam tabel HTML.

2. Form tambah data — create.php
   <input type="tel" name="no_hp" required>
   Digunakan untuk menerima nomor HP mahasiswa sebagai tambahan informasi kontak.

3. Menyimpan data — store.php
   $sql = "INSERT INTO mahasiswa
   (nim, nama, email, no_hp, prodi, angkatan, ipk)
   VALUES (?, ?, ?, ?, ?, ?, ?)";
   Digunakan untuk memasukkan data mahasiswa baru ke database menggunakan prepared statement.

4. Memperbarui data — update.php
   $sql = "UPDATE mahasiswa
   SET nama = ?, email = ?, no_hp = ?, prodi = ?,

```
            angkatan = ?, ipk = ?
        WHERE nim = ?";
```
Digunakan untuk memperbarui informasi mahasiswa berdasarkan NIM.

5. Menghapus data — delete.php
```
$stmt = mysqli_prepare(
    $koneksi,
    "DELETE FROM mahasiswa WHERE nim = ?"
);
```
Digunakan untuk menghapus data berdasarkan NIM dengan prepared statement.

Error yang Pernah Muncul

error:
php : The term 'php' is not recognized as the name of a cmdlet, function, script file, or operable program.

Penyebab:

Windows PowerShell belum mengenali perintah php karena lokasi PHP dari XAMPP belum ditambahkan ke Environment Variable PATH.

Langkah perbaikan:

Program PHP dijalankan menggunakan lokasi lengkap file PHP pada XAMPP, yaitu:

C:\xampp\php\php.exe index.php

Alternatifnya, folder C:\xampp\php dapat ditambahkan ke Environment Variable PATH. Untuk menjalankan aplikasi CRUD melalui browser, Apache dan MySQL harus aktif di XAMPP.